

NANOSTRUCTURE—TRANSPORT PROPERTY RELATIONSHIPS AT ULTRA-HIGH MAGNETIC FIELDS IN HIGH-TEMPERATURE SUPERCONDUCTING TAPES

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Introduction

While the first superconducting materials only operated at very low temperatures (close to absolute zero), the discovery in 1986 of high-temperature superconductors (HTS), such as the cuprate $\text{YBa}_2\text{Cu}_3\text{O}_{7-\delta}$ (YBCO), profoundly renewed the field. These materials enable broader applications thanks to their ability to operate at temperatures achievable with liquid nitrogen (77 K) or to their more robust superconducting properties at low temperatures (20–4 K). HTS tapes, consisting of thin films of RE-BCO materials (RE = rare-earth : Y, Dy, ...) deposited by epitaxy on flexible metallic substrates, are central to the development of energy transport technologies, powerful magnetic devices (MRI magnets, particle accelerators, fusion reactors), and current storage systems [1].

One of the key parameters to control in these applications is the critical current density J_c , i.e., the maximum current density a superconductor can carry without becoming resistive. J_c is highly sensitive to temperature, applied magnetic field, and the material's micro- and nanostructure. In particular, in type-II superconductors such as YBCO, the application of a magnetic field leads to the formation of vortices within the material, and the motion of these vortices carrying a quantum of flux induces dissipation [2]. Since 1986, research and industry have made considerable progress in producing reliable HTS tapes in long lengths (thousands of km/year) with nanostructures optimized to maximize the superconducting critical current (100 kA/mm²) in zero or moderate magnetic fields. One current research focus is to increase J_c under applied magnetic field [3], yet the electrical transport properties of these HTS tapes in high magnetic fields (> 30 T) remain essentially unexplored.

The aim of this internship was to qualify an innovative setup for measuring J_c in intense pulsed magnetic fields, in order to study the influence of YBCO nanostructure on its transport properties (J_c) at various temperatures and up to 60 T. A first series of resistance measurements as a function of temperature and magnetic field (12 T DC) enabled me to extract several characteristic quantities such as the upper critical field and the coherence length, via the Ginzburg–Landau model [4] and the Werthamer–Helfand–Hohenberg–Maki (WHHM) model [5] based on BCS theory [6]. I then performed current–voltage (I–V) measurements using a fast pulsed *FPGA* (Field-Programmable Gate Array) technique, allowing experimental quantification of J_c over a wide range of magnetic fields : in DC field up to 12 T and in pulsed field up to 60 T.

Superconductivity

The superconducting state is characterized by two fundamental properties : zero electrical resistivity and the expulsion of the magnetic field (Meissner effect). The absence of resistance means that a current can flow indefinitely without energy loss. This behavior is linked to the formation of a coherent macroscopic quantum state, in which electrons form Cooper pairs. This coherence prevents scattering from crystal defects, thereby eliminating any dissipation [8, 7].

The BCS theory (Bardeen–Cooper–Schrieffer, 1957) [6] explains this phenomenon through an attractive interaction between electrons, mediated by phonons, leading to the formation of these pairs. They occupy a collective state characterized by a minimal excitation energy : the superconducting gap. While this theory describes conventional superconductors (lead, niobium, etc.) very well, it reaches its limits for HTS such as YBCO, where the pairing mechanism remains uncertain [9]. Nevertheless, several BCS concepts remain relevant, notably the energy gap [10] and the collective nature of the superconducting state.

As early as 1950, Ginzburg and Landau proposed a macroscopic description by introducing a complex wave function $\psi(\vec{r})$ representing the amplitude of the superconducting state [4], as well as two characteristic quantities : the penetration depth λ , which corresponds to the distance over which the magnetic field is expelled inside the superconductor, and the coherence length ξ , which measures the size of spatial variations of the wave function. The interaction with a magnetic field increases the system's free energy ; to minimize it, currents spontaneously appear, generating an opposing field (Meissner effect). When the applied field becomes too strong, two regimes emerge : for $\kappa = \lambda/\xi < 1/\sqrt{2}$, a sharp transition beyond a critical field H_c to the normal state (type I) ; for $\kappa > 1/\sqrt{2}$, the system enters a mixed state between two thresholds H_{c1} and H_{c2} (type II).

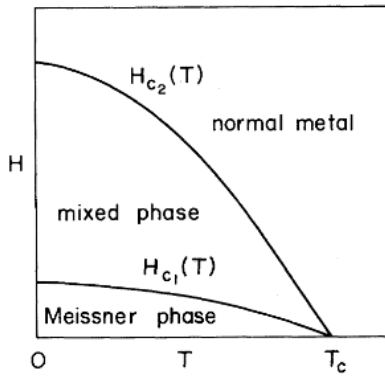


FIGURE 1 – Phase diagram of a type II superconductor as a function of temperature and magnetic field. Three phases are distinguished : the Meissner phase (bottom left) where the field is completely expelled, the mixed phase between $H_{c1}(T)$ and $H_{c2}(T)$ where vortex penetration into the material occurs, and the normal phase above $H_{c2}(T)$ where superconductivity is destroyed. The curves $H_{c1}(T)$ and $H_{c2}(T)$ respectively mark the onset and disappearance of vortices.

Physics of superconducting vortices

In the mixed state, the magnetic field penetrates the material in the form of flux vortices. Each vortex is a stable topological structure carrying a single quantum of flux $\Phi_0 = h/2e$, where the order parameter is zero at its center (of size ξ , the coherence length) [12]. Around this core, superconducting currents circulate to confine the magnetic field, which decays over a typical distance λ (the penetration depth). Vortices interact with each other through repulsion and can form an Abrikosov lattice [13]. Their density is proportional to the local magnetic field.

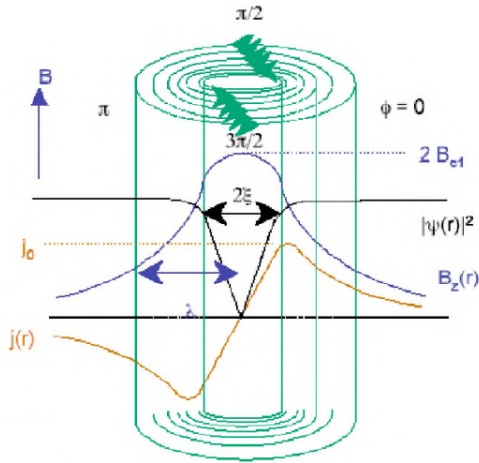


FIGURE 2 – Schematic representation of a quantum vortex in cylindrical geometry [11]. The external magnetic field B (in green) induces a quantized flux trapped at the vortex core, surrounded by circulating currents $j(r)$ (orange curve) that extend over a characteristic distance called the penetration depth λ . The modulus of the superconducting order parameter $|\psi(r)|^2$ (black curve) vanishes at the vortex core and recovers its maximum value beyond the coherence length ξ . The profile of the local magnetic field $B_z(r)$ (blue curve) is centered on the vortex and decays exponentially over the scale λ . The figure also illustrates the flux quantization Φ_0 and the phase structure of the order parameter (green lines), which winds by 2π around the vortex.

Under the action of a current, a Lorentz force $\vec{F}_L = \vec{J} \times \vec{\Phi}_0$ acts on the vortices, tending to move them. Their motion induces dissipation in the metallic core of the vortices, leading to a non-zero resistivity known as the Bardeen–Stephen resistivity [7]. The critical current J_c then corresponds to the threshold above which vortices move collectively, triggering dissipation [14]. This threshold is practically defined by a conventional criterion (e.g., industrial threshold $E_c = 1 \mu\text{V}/\text{cm}$), as well as by the exponent n describing the sharpness of the transition in the I–V curves.

To preserve the dissipation-free state, it is essential to prevent vortex motion through *vortex pinning*. This relies on the presence of defects that stabilize vortices by minimizing their energy. Pinning is more effective at low temperature and low field, and becomes more difficult at high temperature or high magnetic field, when thermal fluctuations or vortex–vortex interactions dominate. In the REBCO thin films used for HTS tapes, advances in epitaxial growth techniques make it possible to add and engineer defects (planar, columnar, nanoparticles, etc.) in order to optimize vortex pinning under different magnetic field and temperature conditions [1, 3]. We refer to Appendix .5 for more details.

YBCO : structure and control of superconductivity

$\text{YBa}_2\text{Cu}_3\text{O}_{7-\delta}$ (YBCO) is a high-critical-temperature superconductor belonging to the cuprate family, with an orthorhombic structure composed of superconducting CuO_2 planes separated by layers containing barium, yttrium, and Cu–O chains, as shown in Fig. 3. These chains act as charge reservoirs, controlling the doping level through oxygenation (δ), with an optimal critical temperature of approximately 92 K for $\delta \approx 0.07$.

Superconductivity emerges within the CuO_2 planes, where the formation of Cooper pairs is confined. This quasi-two-dimensional nature results in strong electronic and magnetic anisotropy, with a relatively weak interlayer coupling [2]. The properties of YBCO (critical temperature around 90 K, large energy gap, high $H_{c2}(0) \approx 120$ T) make it a benchmark material for industrial applications in extreme-field environments (MRI, fusion). From a technological standpoint, YBCO is integrated into so-called REBCO (Rare Earth Ba Cu O) superconducting tapes, deposited as thin films on flexible substrates. Further details are provided in Appendix .4.

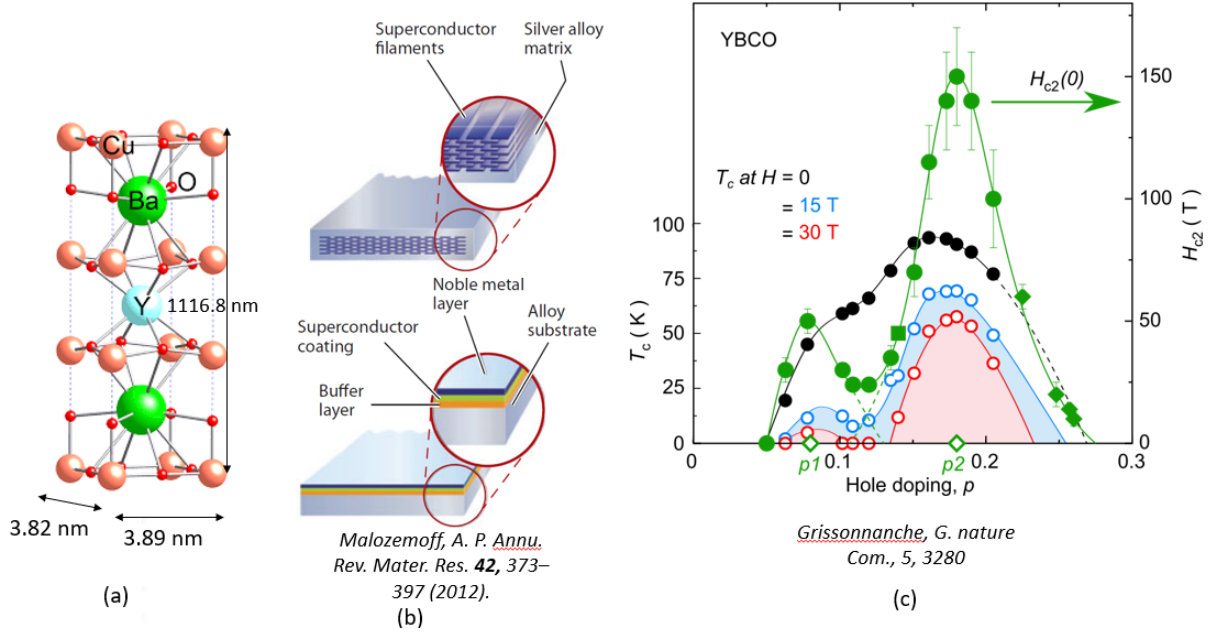


FIGURE 3 – (a) Orthorhombic crystal structure of YBCO. Copper atoms (light orange) and oxygen atoms (red) form the CuO_2 planes responsible for superconductivity, as well as Cu–O chains acting as charge reservoirs. Yttrium (blue) is inserted between two CuO_2 layers, while barium (green) separates the CuO_2 –Y– CuO_2 blocks. (b) Structure of a first- and second-generation REBCO superconducting tape, shown respectively at the top and bottom. (c) The black, blue, and red curves represent the critical temperature as a function of YBCO doping at 0, 15, and 30 T, respectively. H_{c2} as a function of doping is shown in green.

Characterization by $R(T)$ measurements under magnetic field

.1 Samples and Experimental Setup

I participated in resistance–temperature measurements on a series of nine samples provided by the company *CERACO*. These were thin YBCO films deposited on a substrate, patterned in a meander geometry (Fig. 4) to maximize the electrical path length while minimizing the open-loop area. This configuration is particularly well suited for pulsed-field measurements, as it reduces parasitic induction effects while adapting to the spatial constraints of the cryostat.

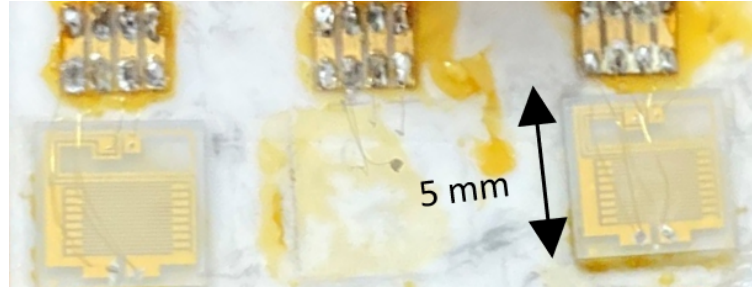


FIGURE 4 – Optical view of two samples, connected by gold wires fixed to the pads with silver lacquer, ensuring stable ohmic contact at low temperature.

All samples were covered with a gold layer, providing protection against deoxygenation and carbonation while ensuring good electrical contact quality. Seven samples featured a bridge width of $40\ \mu\text{m}$, and two samples had a width of $20\ \mu\text{m}$. The latter case is more favorable for pulsed-field measurements [16], although it is also more sensitive to degradation of superconducting properties during etching.

The measurements were performed in an *ICE Oxford* cryostat equipped with a superconducting magnet capable of applying a static magnetic field up to 12 T, perpendicular to the sample plane. To capture fine variations in resistance near the transition, a *Lock-in* detection was used in a four-point configuration, separating current and voltage electrodes. This geometry eliminates parasitic contributions from contact resistances, ensuring measurement fidelity. Further details are given in Appendix .1.

The *Lock-in* amplifier allows sensitive detection of weak signals buried in noise. A sinusoidal alternating current is injected, and the measured voltage is correlated with a reference at the same frequency. Only the synchronous component is retained, removing out-of-band noise (thermal noise, interference). This setup is therefore ideal for tracking the resistance drop over several orders of magnitude, typical of the transition region. Measurements were carried out between 65 K and 90 K for several magnetic fields ranging from 0 to 12 T.

.2 Phase Transition Analysis

From the $R(T)$ curves obtained for each sample at different magnetic fields, I was able to extract several observables : the critical temperature T_c , the irreversibility field H_{irr} (field at

which vortices start to move), and the upper critical field H_{c2} as a function of temperature. The latter two are particularly important for describing the respective limits of macroscopic superconductivity and the survival of Cooper pairs under an applied magnetic field.

The measured transitions all have a finite width due to thermal fluctuations, but a residual resistance tail is also visible here, typical of the presence of microscopic inhomogeneities in the material. This behavior is systematic, regardless of bridge width, confirming an effect linked to fabrication. This may result from local variations in oxygenation or structural defects from the deposition or bridge fabrication process.

To extract H_{irr} and H_{c2} , I use a simple and robust method based on empirical criteria on the $R(T)$ curve. The irreversibility field is defined as the field for which the resistance becomes non-zero, according to a fixed threshold (1% of the normal-state resistance at high field in our case). This field marks the disappearance of the zero-resistance state, often associated with entry into a phase where vortices can move freely. The upper critical field is defined as the field for which the resistance reaches 99% of its value in the normal state, corresponding to the disappearance of the last Cooper pairs (see Fig. 5).

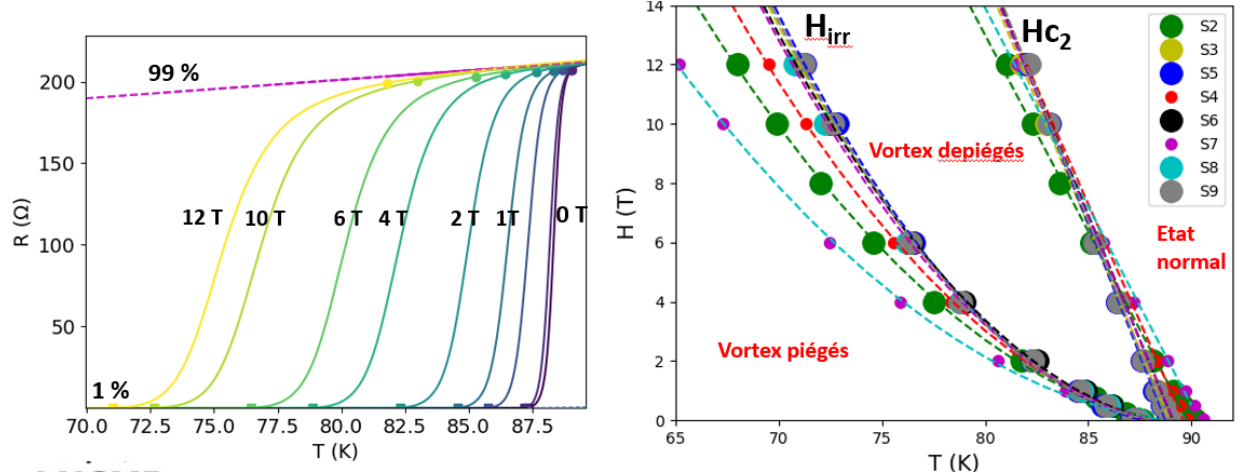


FIGURE 5 – Right : Resistance as a function of temperature between 0 and 12 T. The dashed lines correspond to the 99% and 1% criteria of the normal-state resistance. Left : H -temperature diagram, where large and small dots correspond to 40 μm and 20 μm bridges, respectively. Dashed lines correspond to the linear Ginzburg–Landau fit for H_{c2} and an empirical polynomial fit for H_{irr} .

The evolution of $H_{c2}(T)$ follows a linear decrease near T_c , consistent with the Ginzburg–Landau model [4]. However, this trend should bend at low temperature as predicted by the WHHM model [5] based on BCS theory [6]. In our case, an extrapolation suggests $H_{c2}(0) > 103$ T. Since measurements are limited to a range close to T_c , both models are indistinguishable; see Appendix .2 for further details.

Finally, the reproducibility of the transitions appears to depend on bridge width. Wide bridges show consistent transitions between samples, whereas narrow bridges display more dispersion, suggesting increased sensitivity to local inhomogeneities. At this scale, microscopic inhomogeneities have a proportionally larger impact.

Current–voltage measurement

.1 Experimental setup and acquisition method

In order to study the I–V transport properties of our samples in pulsed magnetic fields (up to 60 T), we first validated the experimental setup by performing measurements in static magnetic fields (up to 12 T).

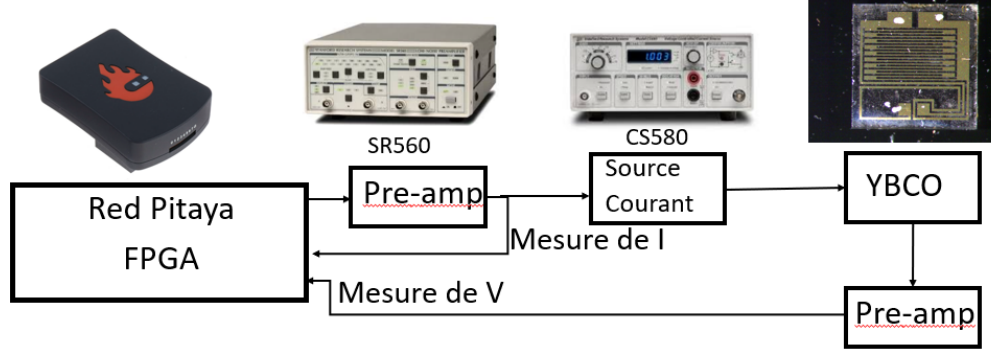


FIGURE 6 – Experimental setup for I–V measurements

The measurement is based on injecting short current pulses ($t = 10 - 500 \mu\text{s}$) into the sample and simultaneously recording the voltage response. The setup is built around a *Red Pitaya*, an *FPGA* platform that generates the pulses, acquires the signals, and integrates an automatic overvoltage protection system, thereby limiting the risk of sample destruction. The *Red Pitaya* includes a data acquisition (DAQ) system with two analog-to-digital converters (ADC) sampling at 125 MHz, and a deterministic programmable hardware logic operating at 140 MHz, which provides a time resolution suited to our few-microsecond pulses and ensures timely reaction when I_c is exceeded.

The *Red Pitaya* sends a ± 1 V pulse, amplified by an *SR560*, then converted into current by a *CS580* source. This current is injected using a four-point configuration. The voltage response is amplified by a second *SR560* and read on input 1 of the *Red Pitaya*. This system enables the plotting of I–V curves for each applied temperature and magnetic field.

I performed the measurements on three wide-bridge samples ($40 \mu\text{m}$), using two protocols : at constant temperature (65 K, 70 K, 75 K) while varying the magnetic field from 0 to 12 T, and at constant field (6 T) while varying the temperature from 65 to 85 K. For each point, current pulses were injected, the raw signals in *TDMS* format were processed using a *Python* script I wrote to extract the raw data, average the pulses, and build the I–V curve.

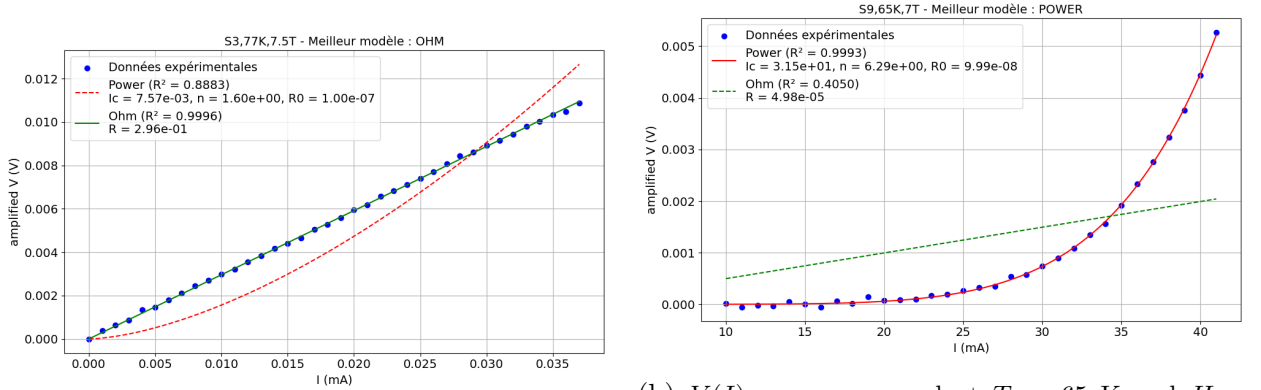
.2 Analysis of Current–Voltage Curves

The $V(I)$ curves reveal two distinct regimes. At low temperature and moderate magnetic field, the response is non-linear, a signature of a regime with pinned vortices. At higher temperature or under strong magnetic field, the vortices move freely and the response becomes

linear. To model these behaviors, two laws are used. In the non-linear regime, an empirical power law, standard in the applied superconductivity community, is fitted as :

$$V(I) = V_0 \left(\frac{I}{I_c} \right)^n + R_{\text{def}} I \quad (1)$$

It contains three important parameters : the n-value, the critical current I_c , and the voltage threshold V_0 that defines I_c and is conventionally set from the electric field at $1 \mu\text{V}/\text{cm}$. Here, I fixed the voltage threshold V_0 to $500 \mu\text{V}$ (i.e., $10 \mu\text{V}/\text{cm}$), which allows defining a critical current I_c close to the standard $1 \mu\text{V}/\text{cm}$ criterion. R_{def} accounts for residual dissipation. When vortices are depinned, the curves become linear and are simply modeled by an Ohmic law : $V(I) = R_{\text{IV}} I$.



(a) $V(I)$ curve measured at $T = 77 \text{ K}$ and $H = 7.5 \text{ T}$, in a regime where vortices are depinned. Blue points correspond to experimental data, the solid curve represents an Ohmic fit $V = R_{\text{IV}} I$, while the dashed curve shows the best possible fit with a power law, which fails to correctly describe the data.

(b) $V(I)$ curve measured at $T = 65 \text{ K}$ and $H = 7 \text{ T}$, in a regime where vortices are mostly pinned. Blue points correspond to experimental data. The fit with a power law $V = V_0(I/I_c)^n + R_{\text{def}} I$ (solid curve) correctly describes the observed non-linear behavior, whereas the Ohmic linear model (dashed) does not capture the critical transition.

FIGURE 7 – Examples of two $V(I)$ curves : in one case vortices are mobile, in the other they are pinned.

The evolution of the critical current I_c (figure 9) shows that it decreases with increasing magnetic field, consistent with a growing vortex density reducing the ability to carry current without dissipation. I_c also decreases with temperature, due to the progressive weakening of superconductivity as T_c is approached.

.3 Measurement in Pulsed Magnetic Field

I then took part in pulsed-field measurements up to 35 T on a sample (S3) at different temperatures : $T = 77 \text{ K}$, $T = 70 \text{ K}$, $T = 65 \text{ K}$, and $T = 50 \text{ K}$. The objective of these measurements is to verify the robustness and reproducibility of our measurement method through direct comparison with static-field measurements.

The measurement protocol remains identical : current pulses are injected into the sample, and the voltage is then measured. These current pulses are synchronized with the magnetic field profile, which varies rapidly, as illustrated in figure 8.

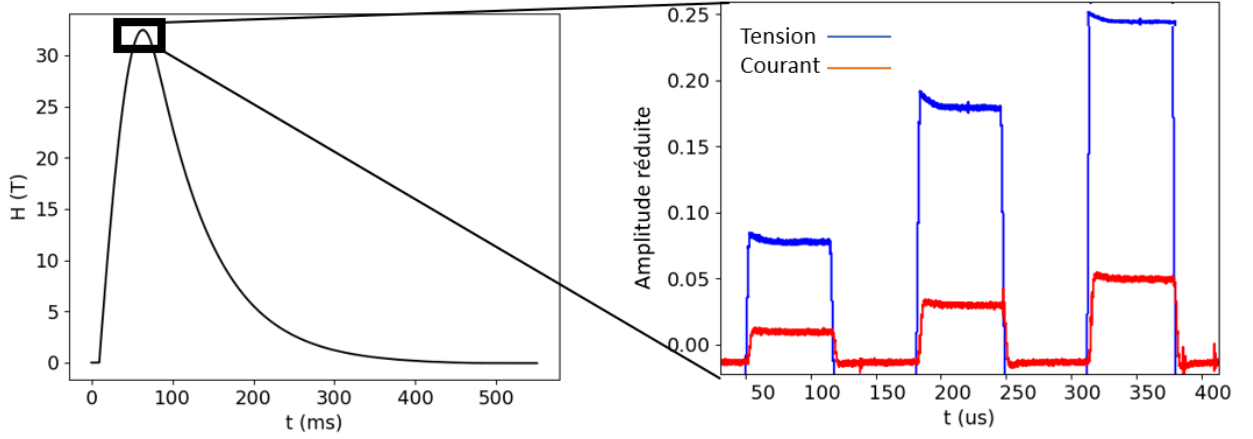


FIGURE 8 – Magnetic field pulse profile at 32 T over 400 ms. Inset : zoom at the field peak showing a series of I-V pulses in the sample over approximately 400 μ s.

The results for the critical current as a function of magnetic field are shown in figure 9, after processing with Python code that I wrote. In the overlapping field range (6–12 T), the data show good overall reproducibility, thus validating the pulsed-field measurement. However, the I_c values obtained in pulsed fields are lower than those measured in static fields. This difference could be due either to slight degradation of the sample between the two measurement series, or to greater uncertainty in the actual sample temperature in the pulsed configuration—where thermal contact is poorer due to the use of a plastic sample holder, unlike the copper holder used for static-field measurements. Additionally, the field pulse itself may induce heating in the sample.

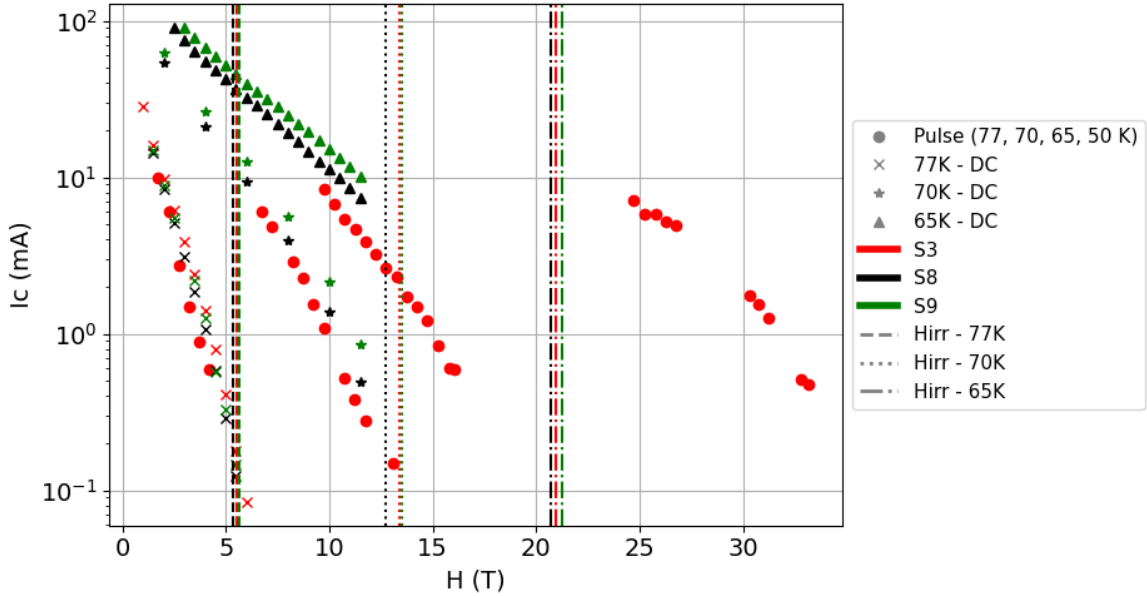


FIGURE 9 – Comparison of critical currents as a function of magnetic field obtained in static and pulsed fields at different temperatures.

Conclusion

This work made it possible to characterize the superconducting transport properties of thin-film YBCO samples by combining measurements in static magnetic fields up to 12 T and in pulsed fields up to 60 T. The $R(T)$ curves allowed us to extract the H – T phase diagram, including the H_{c2} and H_{irr} fields. The $I(V)$ measurements revealed the different regimes of vortex dynamics, whether pinned or mobile. Finally, the first experiments in pulsed magnetic fields validated our measurement protocol, showing good reproducibility with the static-field measurements, despite a slight decrease in I_c . With this measurement protocol now validated, we can continue exploring the influence of nanostructure on critical currents.

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ANNEXES

.1 Experimental setups



FIGURE 10 – Experimental setups in static field (left) and pulsed field (right).

The static-field experimental setup is a closed-cycle ^4He cryogenic system equipped with a Gifford–McMahon (GM, cryofree) cold head from ICE Oxford. The base temperature of the variable temperature insert (VTI) is 1.5 K, and the magnet generates a magnetic field ranging from 0 to 12 T.

For pulsed fields, the experiment is conducted in a shielded room (box 14) equipped with a 60 T pulsed magnet, along with dedicated instrumentation racks for lock-in resistivity measurements and pulsed I–V measurements.

.2 Superconductivity

Ginzburg-Landau Theory

As discussed in the main report, the Ginzburg-Landau theory is based on a continuous thermodynamic description of the superconducting state [4], via a complex order parameter $\psi(r)$, representing the amplitude of the superconducting state at the macroscopic scale. This order parameter does not describe a quantum wavefunction in the strict sense, but rather the density of Cooper pairs in the material ($|\psi|^2$). This theory does not depend on the microscopic pairing mechanism, making it a particularly powerful tool for describing unconventional superconductors [9] such as YBCO.

The free energy of the system is expanded as a function of the order parameter and its coupling to the magnetic field via the vector potential $A(r)$, and is written as [7] :

$$\mathcal{F}[\psi, A] = \int d^3r \left[\alpha |\psi|^2 + \frac{\beta}{2} |\psi|^4 + \frac{1}{2m^*} |(-i\hbar\nabla - 2eA)\psi|^2 + \frac{B^2}{2\mu_0} \right], \quad (2)$$

where α depends on temperature and changes sign at the transition, β is positive to ensure stability, and m^* is the effective mass of the Cooper pairs.

Minimizing the free energy leads to two fundamental equations : a nonlinear Schrödinger equation and an equation for the superconducting current :

$$\begin{aligned} \frac{1}{2m^*} (-i\hbar\nabla - 2eA)^2 \psi + \alpha\psi + \beta|\psi|^2\psi &= 0 \\ j_s &= \frac{2e\hbar}{m^*} \mathcal{Im} \left\{ \psi^* \left(\nabla - \frac{2ie}{\hbar} A \right) \psi \right\} \end{aligned} \quad (3)$$

The first equation can be interpreted as a stationary nonlinear Schrödinger equation for a particle of mass m^* , coupled to an electromagnetic field, with an effective potential $V_{\text{eff}} = \alpha|\psi|^2 + \frac{\beta}{2}|\psi|^4$. This potential has a global minimum at $\psi = 0$ if $\alpha > 0$ and $|\psi|^2 = -\alpha/\beta$ if $\alpha < 0$. The emergence of a nonzero minimum for ψ when $T < T_c$ corresponds to the spontaneous breaking of U(1) symmetry. The system chooses a global phase θ (which becomes a real physical variable), enabling topological phenomena such as vortices [13]. Indeed, in a vortex, the phase winds around a point according to :

$$\oint \nabla\theta \cdot d\vec{l} = 2n\pi \quad (4)$$

This implies that ψ must vanish at the vortex core for its phase to be well-defined, creating a locally normal region.

As the magnetic field increases, the superconducting state becomes unstable. The upper critical field H_{c2} corresponds to the field beyond which superconductivity vanishes completely : $\psi = 0$ everywhere.

Near H_{c2} , the order parameter is small, allowing the Ginzburg–Landau equation to be linearized by neglecting the nonlinear $|\psi|^2$ term :

$$\frac{1}{2m^*} (-i\hbar\nabla - 2e\vec{A})^2 \psi + \alpha\psi = 0 \quad (5)$$

This equation is formally equivalent to a Schrödinger equation for a charged particle in a magnetic field, with an effective quadratic potential. Its solutions are Landau levels, with quantized energies. The condition for the disappearance of superconductivity corresponds to the moment when the lowest Landau level energy equals $-\alpha$ (the effective potential).

The ground-state energy of an electron of charge $q = -2e$ in a field B is :

$$E_0 = \hbar\omega_c/2 = \frac{\hbar|q|B}{2m_e} \quad (6)$$

By imposing $E_0 = -\alpha$, we find the critical value $H_{c2} = -2\alpha m_e/(\hbar e)$. To connect this to measurable quantities, we introduce the coherence length defined by :

$$\xi(T)^2 = -\frac{\hbar^2}{4m_e\alpha(T)} \quad (7)$$

This finally yields the canonical expression $H_{c2}(T) = \Phi_0/2\pi\xi(T)^2$, with $\Phi_0 = \frac{h}{2e}$ the magnetic flux quantum. Near T_c , the parameter $\alpha(T)$ varies linearly, which implies that the upper critical field close to T_c can be expressed as :

$$H_{c2}(T) = H_{c2}(0) \left(\frac{T_c - T}{T_c} \right) \quad (8)$$

In our case, we can compute $\xi(T)$ from our $H_{c2}(T)$ data, then fit it with a GL-like model such as $\xi(T) = \xi(0) (1 - T/T_c)^{-\alpha}$. The parameters $\xi(0)$, T_c , and α were freely adjusted to the experimental data, yielding $\xi(0) = 1.35$ nm, $T_c = 89.09$ K, and $\alpha = 0.54$. The value of $\xi(0)$ is consistent with expectations for YBCO, indicating a very short coherence length, typical of cuprates. Finally, the exponent α is close to the expected value in Ginzburg–Landau theory ($\alpha^{\text{GL}} = 0.5$).

Bardeen–Cooper–Schrieffer (BCS) Theory

As discussed in the main report, the BCS theory is based on the formation of Cooper pairs [14], which are attractive correlations between electrons, leading to a coherent macroscopic quantum state.

The system Hamiltonian is written as the sum of free electrons and an attractive interaction between pairs :

$$\mathcal{H} = \sum_{\vec{k}\sigma} \epsilon_{\vec{k}} c_{\vec{k}\sigma}^\dagger c_{\vec{k}\sigma} + \sum_{\vec{k}\vec{k}'} V_{\vec{k}\vec{k}'} c_{\vec{k}\uparrow}^\dagger c_{-\vec{k}\downarrow}^\dagger c_{\vec{k}'\downarrow} c_{\vec{k}'\uparrow} , \quad (9)$$

where $\epsilon_{\vec{k}}$ is the energy of a free electron relative to the Fermi level, $c_{\vec{k},\sigma}^\dagger$ and $c_{\vec{k},\sigma}$ are electron creation and annihilation operators, and $V_{\vec{k},\vec{k}'}$ is the attractive interaction, limited to states near the Fermi level due to electron–phonon coupling.

Cooper showed that a two-electron state above the Fermi sea, with attractive interaction, is unstable and forms a bound state. The ground state generalizes this idea to an entire condensate of pairs : each pair consists of two electrons with opposite momenta $\vec{k}$, $\vec{k}'$ and opposite spins $\uparrow$, $\downarrow$, resulting in a symmetric spatial wavefunction (spin-singlet state) :

$$|\text{BCS}\rangle = \prod_{\vec{k}} \left(u_{\vec{k}} + v_{\vec{k}} c_{\vec{k}\uparrow}^\dagger c_{-\vec{k}\downarrow}^\dagger \right) |0\rangle , \quad (10)$$

where $u_{\vec{k}}^2$ and $v_{\vec{k}}^2$ represent the probabilities of finding a $\vec{k}$ state empty or occupied by a pair.

By minimizing the system's energy, we obtain two equations. The first is the self-consistent gap equation [14, 8] :

$$\Delta_{\vec{k}} = - \sum_{\vec{k}'} V_{\vec{k}\vec{k}'} \frac{\Delta_{\vec{k}'}}{2E_{\vec{k}'}} \tanh \left(\frac{E_{\vec{k}'}}{2k_B T} \right) , \quad (11)$$

with $\Delta_{\vec{k}}$ being the superconducting gap amplitude, which plays a role analogous to the Ginzburg–Landau order parameter but at the microscopic scale.

The second equation is the electronic density of states, which becomes :

$$N(E) = N(0) \frac{E}{\sqrt{E^2 - \Delta^2}} \quad (12)$$

This equation shows a divergence at $E = \Delta$, indicating the existence of an energy gap for exciting quasiparticles. This explains the zero resistance and strong coherence at low temperatures.

Werthamer–Helfand–Hohenberg–Maki Model

The WHHM model [5] describes the evolution of the upper critical field $H_{c2}(T)$ as a function of temperature, incorporating :

- Orbital effect : Under a magnetic field, electrons describe quantized orbits (Landau levels), progressively destroying the coherence of Cooper pairs. This effect dominates near T_c .
- Pauli effect : A magnetic field aligns spins, which breaks apart the opposite-spin electrons forming a Cooper pair. This effect becomes important at low temperatures.
- Spin–orbit coupling : Acts as a “spin mixer” that limits the Pauli effect by preventing spins from aligning too efficiently with the field. It can stabilize superconductivity against the magnetic field.
- Maki parameter α [A.1] : Quantifies the competition between orbital and Pauli effects.

This model is based on a microscopic solution of the BCS equations in a magnetic field. Unlike the Ginzburg–Landau model, it remains valid at temperatures well below T_c , as long as superconductivity persists.

The WHHM model is applied here in the dirty limit, where the mean free path is much shorter than the coherence length. This assumption is often valid for real superconductors, particularly thin films. In this regime, electron scattering modifies the magnetic field response, affecting the shape of $H_{c2}(T)$. The critical field is obtained as the solution to the equation :

$$\begin{aligned} \ln \left(\frac{1}{t} \right) = & \left(\frac{1}{2} + \frac{i\lambda_{SO}}{4\gamma} \right) \psi \left(\frac{1}{2} + \frac{\tilde{h} + \lambda_{SO}/2 + i\gamma}{2t} \right) + \\ & + \left(\frac{1}{2} - \frac{i\lambda_{SO}}{4\gamma} \right) \psi \left(\frac{1}{2} + \frac{\tilde{h} + \lambda_{SO}/2 - i\gamma}{2t} \right) - \psi \left(\frac{1}{2} \right), \end{aligned} \quad (13)$$

where $t = T/T_c$ is the reduced temperature, $\tilde{h} = 4H_{c2}/(\pi^2 T_c |dH_{c2}/dT| T = T_c)$ is the reduced magnetic field, λ_{SO} is the spin–orbit coupling parameter, α is the Maki parameter quantifying the Pauli effect, $\gamma = \sqrt{(\alpha\tilde{h})^2 - (\lambda_{SO}/2)^2}$, and $\psi(x)$ is the digamma function.

At temperatures close to T_c , the orbital effect dominates, and the WHHM solution reduces to Ginzburg–Landau theory, while at low temperatures, the Pauli effect takes over, and $H_{c2}(T)$ saturates at a limiting value set by pair breaking due to spin alignment. A nonzero spin–orbit coupling increases this limit.

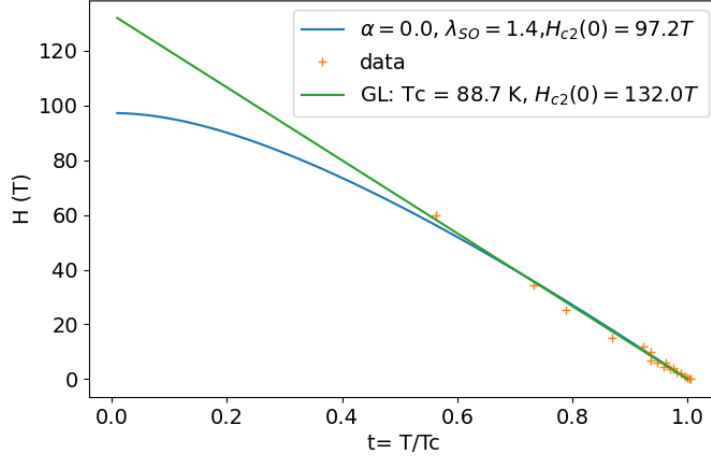


FIGURE 11 – Upper critical field as a function of reduced temperature (orange crosses) compared to the Ginzburg–Landau model (solid red line) and the WHHM model (solid blue line).

This model allows for a more accurate extrapolation of $H_{c2}(0)$ than the Ginzburg–Landau model, while also identifying the nature of the physical limitations.

The Ginzburg–Landau fit extrapolates to $H_{c2}(0) \approx 132.0$ T. In reality, H_{c2} will saturate at low temperature, for instance with the WHHM model, where saturation is mostly driven by α . In our case, I implemented the WHHM fit, but more low-temperature data points would be needed to study the WHHM fit accurately and extract meaningful parameter values.

.3 Phase transitions : vortex glass, vortex liquid

At zero temperature and in the presence of a regular array of defects, vortices tend to arrange themselves into a crystalline lattice (Abrikosov lattice) [13]. In reality, however, superconductors are subject to thermal fluctuations, high magnetic fields, and varying degrees of structural disorder. These factors can cause the vortex lattice to melt, leading to disordered phases analogous to those observed in conventional condensed matter : liquid and glass.

When thermal fluctuations become sufficiently strong, the Abrikosov lattice can lose its topological coherence. Vortices then cease to be localized on an ordered lattice and enter a fluid-like state called the vortex liquid. In this phase, vortices are free to move, which induces a finite resistivity as soon as a current is applied, even if the current is very small. Conversely, at low temperature and in the presence of sufficient disorder, vortices can organize into a disordered but frozen state called the vortex glass [A.2]. In this phase, vortices are collectively pinned by the defect landscape but without long-range order. This is a non-crystalline state characterized by a nonlinear response to current excitation, the absence of dissipation at zero current, and long relaxation times.

Even in the vortex glass state, thermally activated relaxation processes can occur. This phenomenon, called flux creep [A.4], corresponds to the intermittent motion of vortices over long timescales, due to their ability to overcome pinning energy barriers under the influence of temperature. This behavior results in residual dissipation, even below the critical current,

and can be modeled by a law of the form :

$$E(J) \propto \exp\left(-\frac{U(J)}{k_B T}\right),$$

where $U(J)$ is the pinning energy barrier, which decreases with current. For a vortex glass model, this barrier can depend on the current as $U(J) \approx J^{-\mu}$. Flux creep is a limiting factor for applications : even without exceeding J_c , HTS superconductors can exhibit small but nonzero losses.

In high- T_c materials such as YBCO, these transitions are particularly pronounced due to strong anisotropic coupling, significant thermal fluctuations, and the ability to introduce controlled defects [2]. These transitions are crucial for applications, as they determine dissipation stability performance.

.4 HTS tapes

Since the discovery of high-critical-temperature superconductors, extensive research has focused on potential industrial applications. The first industrial developments began with the production of Bi-2223 tapes (bismuth, strontium, calcium, copper oxide), but these tapes showed limitations in terms of mechanical performance and cost [A.5]. Subsequently, YBCO-based tapes, known as second-generation (2G) tapes, were introduced in the 2000s [A.6]. REBCO tapes (Rare Earth Barium Copper Oxide), in which the yttrium in YBCO is partially or entirely replaced by a rare earth element (Gd, Sm, Nd), are today the most technologically advanced for applications requiring high current densities in extreme environments (fusion, accelerators, etc.). They generally consist of several functional layers deposited on a metallic substrate (often Ni-W or Hastelloy).

Their architecture, composed of a metallic substrate (thickness about 50–100 μm), provides mechanical flexibility, robustness, and thermal conduction ; buffer layers (often MgO, CeO₂) which ensure chemical isolation from the substrate and allow biaxial texture (alignment of the crystal lattice along two directions) ; an epitaxially grown superconducting REBCO layer of about 1 μm thickness ; and a stabilizing layer, typically 1 to 2 μm of silver followed by a copper plating (50 μm) to ensure thermal dissipation in case of quench and to protect the superconductor.

REBCO tapes offer several advantages, such as a high critical current density ($J_c > 10^{10}$ A/m²) maintained under high magnetic fields (30–40 T at 4 K) [A.7]. They can also operate at higher temperatures (20–77 K), reducing cryogenic costs, particularly compared to conventional superconductors such as NbTi and Nb₃Sn, which require temperatures close to liquid helium and are limited to fields below 15–30 T. However, their production remains costly, largely due to the complexity of thin-film deposition processes and the need for highly precise crystalline texturing. Moreover, the growth of the superconducting film is sensitive to defects in the underlying buffer layers, which can cause local drops in critical current or non-superconducting regions.

REBCO tapes have already found applications, for example in superconducting magnets capable of generating magnetic fields above 20 T, used in nuclear fusion projects and in certain ultra-high-field nuclear magnetic resonance systems (1.3 GHz at 30.5 T).

They are also used in superconducting cables for dense urban power grids, where their ability to carry currents of several kiloamperes increases grid capacity without increasing footprint.

.5 Defects in REBCO superconductors and their role in vortex pinning

Nature of defects in REBCO layers

During the growth of thin films on a substrate, various types of defects may appear, influencing the transport properties. These defects may be intrinsic (related to the film growth process) or extrinsic (intentionally introduced through defect engineering).

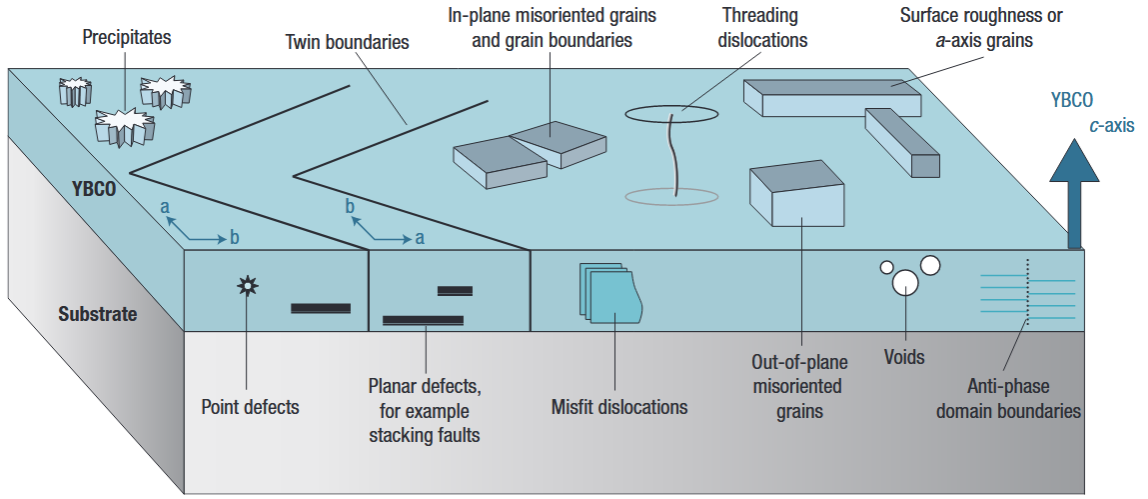


FIGURE 12 – Different types of defects proposed for vortex pinning [1].

Among the most common intrinsic defects, one can distinguish :

- Dislocations and vertical growth defects [A.8] : the deposition process may create vertical defect lines, often arising from slight misalignments between crystal grains or sudden changes in the film growth mode.
- Planar defects [A.8] : flat, extended defects such as microcracks, irregular stacking of atoms, or non-superconducting layers trapped between superconducting layers.
- Composition or oxygenation inhomogeneities [A.11] : as discussed in the YBCO section, superconducting properties depend on the oxygen doping of the material. If certain areas of the tape contain less oxygen, regions with reduced critical temperature or critical current may occur.

Among the most common extrinsic defects (also called artificial pinning centers, APC), one can distinguish :

- Secondary nanoparticles [A.9] : the intentional addition of non-superconducting particles a few nanometers in diameter (e.g., BaZrO_3 , BaHfO_3) into the REBCO layer.
- Columnar defects created by irradiation [A.10] : the tapes can be modified after fabrication by irradiation with highly energetic particles (a few GeV), creating long amorphous columns a few nanometers in diameter.

- Planar defects [A.8] : flat, extended defects such as microcracks, irregular stacking of atoms, or non-superconducting layers trapped between superconducting layers.
- Composition or oxygenation inhomogeneities [A.11] : as discussed in the YBCO section, superconducting properties depend on the oxygen doping of the material. If certain areas of the tape contain less oxygen, regions with reduced critical temperature or critical current may occur.

The critical current density is directly related to the material's ability to pin vortices. However, an excess of defects or a poor distribution can locally suppress the superconducting gap. There is therefore a trade-off between effective pinning and the preservation of superconducting coherence.

The Dew-Hughes model : classification and pinning efficiency

As discussed, the behavior of the critical current as a function of magnetic field directly depends on the nature and efficiency of the defects. To describe this phenomenon, the Dew-Hughes model (1974) [A.12] proposes an empirical approach.

The model starts from the definition of the volumetric pinning force, given by :

$$F_p(H) = J_c(H)\mu_0 H \quad (14)$$

Dew-Hughes then proposes an empirical form for this curve through the relation $f_p(h) \propto h^p(1-h)^q$, where $f_p = F_p/F_{p,\max}$ is the reduced pinning force, $h = H/H_{c2}$, and p, q depend on the dominant type of defect. In our study, we extracted the $F_p(H)$ curve from $J_c(H)$ measurements and fitted it using the Dew-Hughes function.

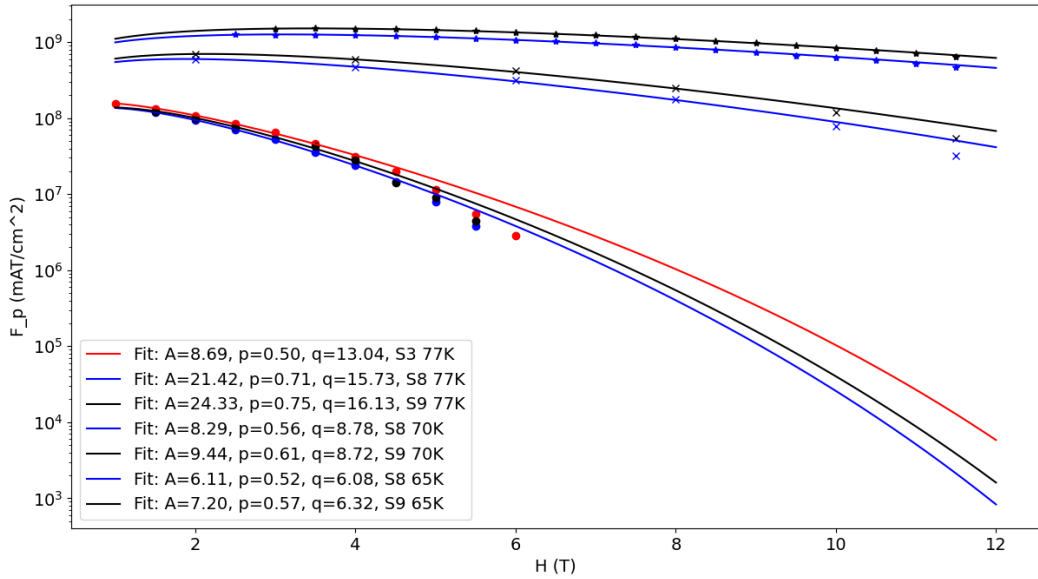


FIGURE 13 – Pinning force curves $F_p(H)$ at different temperatures (77 K, 70 K, 65 K) for several samples (points), fitted using the Dew-Hughes model (solid line).

The values from our fit indicate a pinning force maximum very close to zero field :

$$\frac{df_p(h)}{dh} = 0 \Rightarrow h_{\max} = \frac{p}{p+q} \quad (15)$$

This suggests that the dominant defects at high temperature are weak-link types, such as vertical dislocations, planar defects, or microcracks. We observe that the pinning efficiency improves as the temperature decreases : the value of q decreases, indicating better high-field critical current retention. This may be due to the reduced thermal activation of vortices at lower temperatures.

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